rebound.

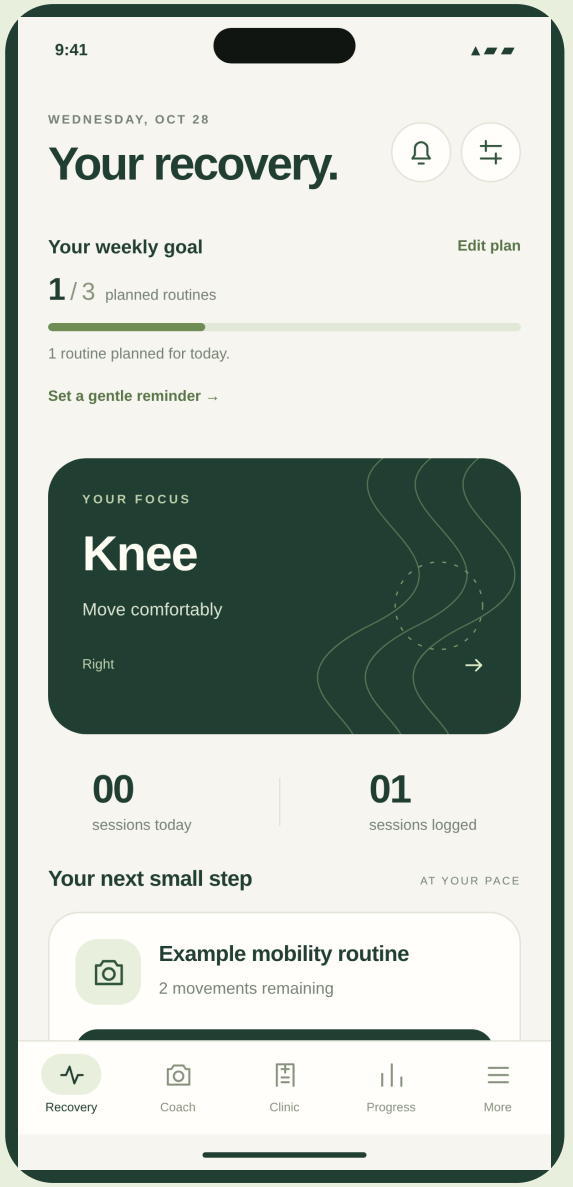
CONGRESSIONAL APP CHALLENGE

Small steps. A steadier routine.

A recovery companion for athletes and everyday people.

Ritvik Saidapur
Siddharth Senthilkumar

North Carolina's 4th District · NC-04



Native UI preview · example data

Recovery needs more than instructions.

Between appointments, a patient still needs clarity, confidence, and consistency.

01

CLARITY

What is my plan today?

A routine should be easy to find and understand.

02

CONFIDENCE

Am I moving as intended?

Instructions need a useful visual reference.

03

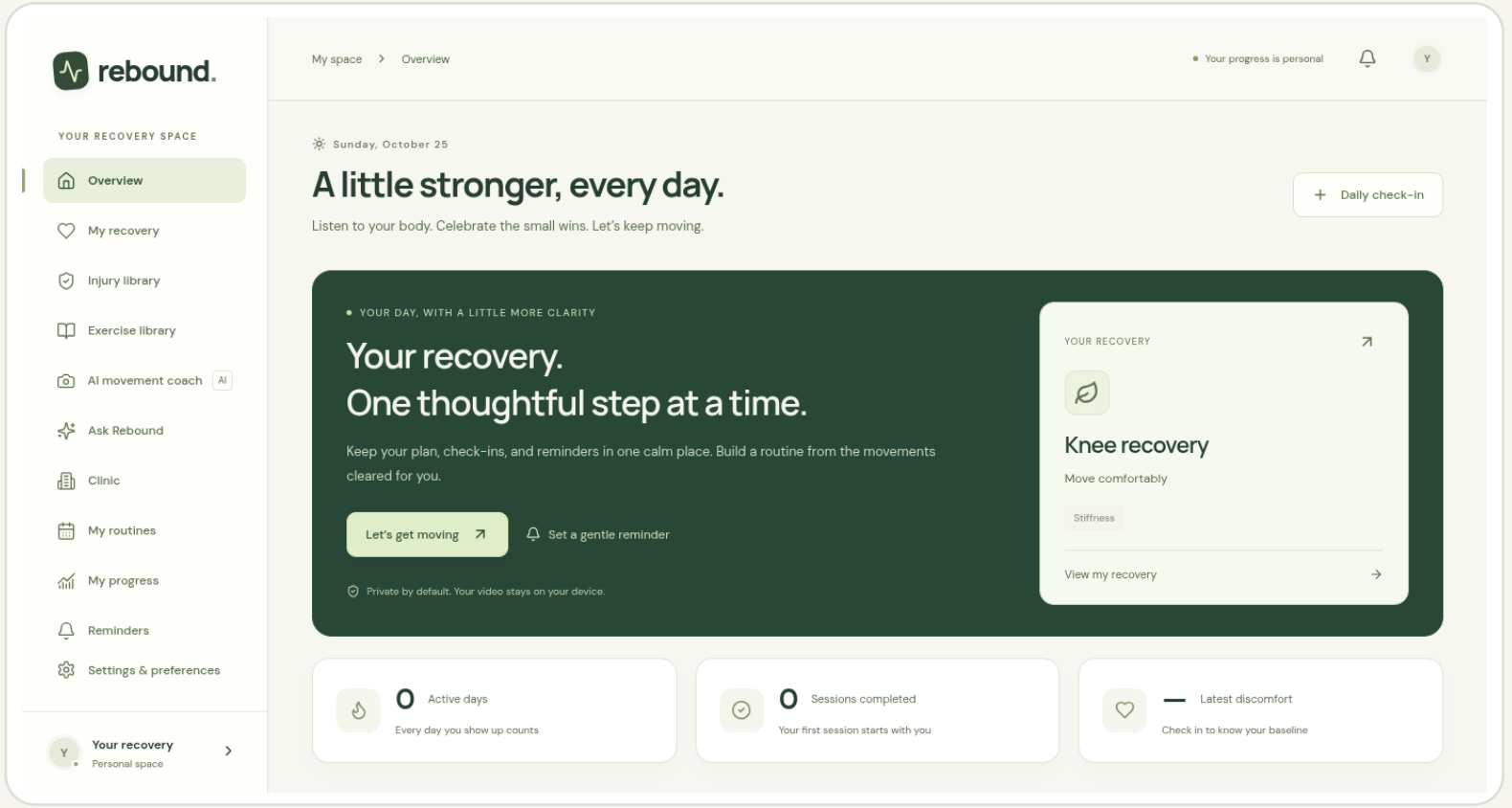
CONSISTENCY

How will I remember?

The next step should fit into an ordinary week.

One calm place for the next step.

Rebound brings the plan, practice, and personal record together.



01 **Plan**

Know what is scheduled.

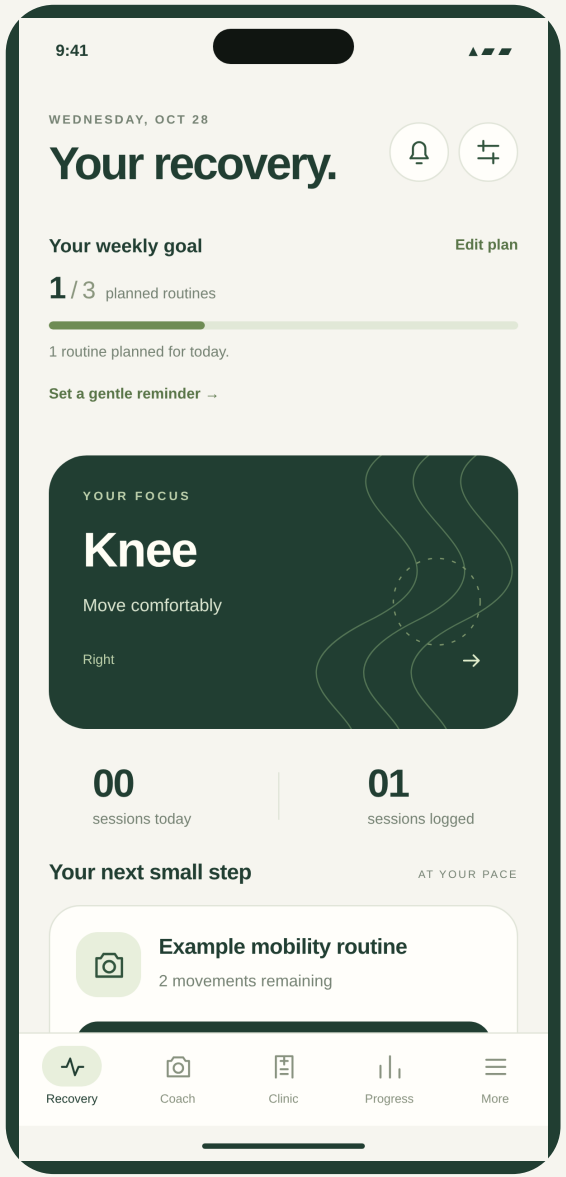
02 **Practice**

See the movement and work at your pace.

03 **Reflect**

Log a session and prepare for a check-in.

Website interface capture · no patient data



CONSISTENCY, WITH ROOM TO REST

A week you can follow.

01

A meaningful weekly goal

Count completed routines against the days in your plan.

02

A timely nudge

Choose reminder days and a time that fits your routine.

03

Rest belongs in the plan

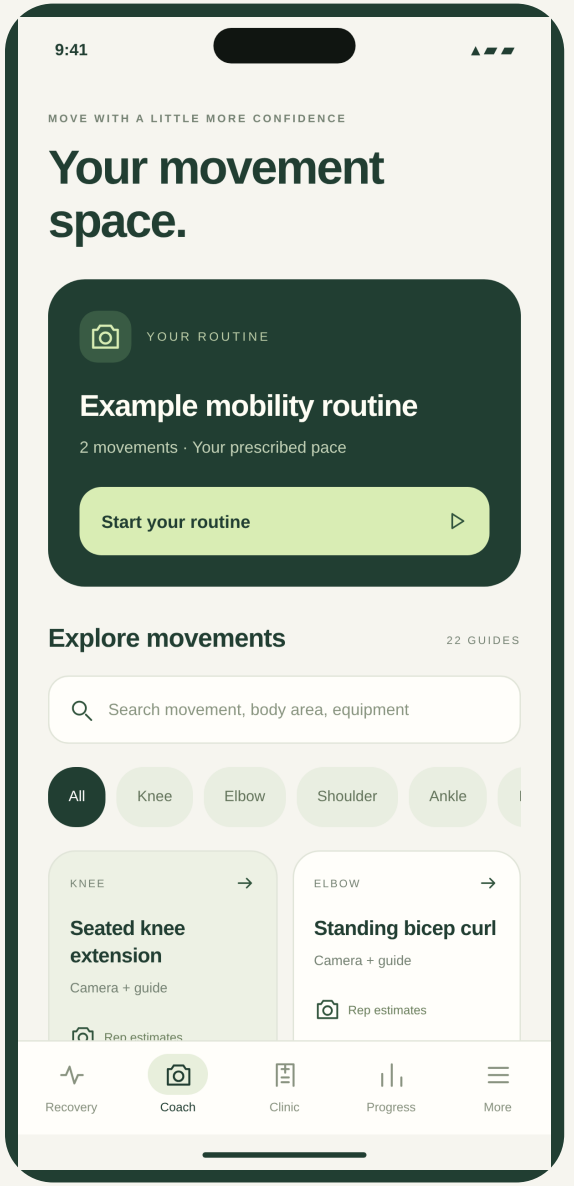
Extra sets do not earn extra goal credit.

Learn the movement. Then practice.

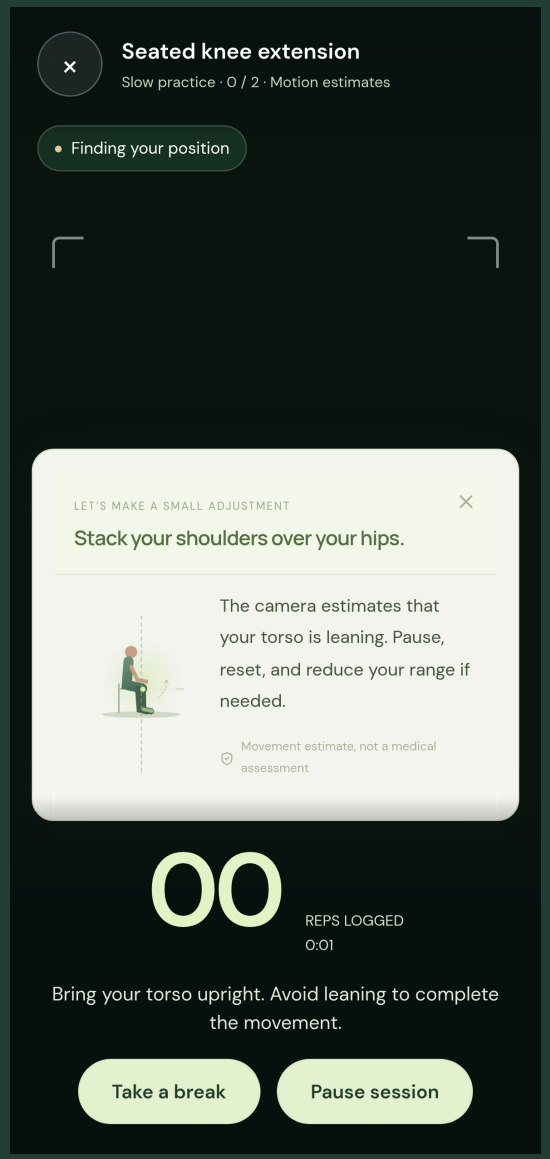
Start the routine first.
Explore the library when you need it.

22 illustrated movement guides

Step-by-step teaching
Slow practice for supported camera movements
Band and dowel examples with individual clearance



Native UI preview · example data · exercise choice and dose require individual clearance



ON-DEVICE CAMERA COACH

Guidance where you are looking.

- **A full-screen view**
  A large rep counter and simple controls.
- **A timely correction cue**
  An illustrated adjustment appears inside the camera view.
- **Pause, breathe, resume**
  Stillness pauses the session; set breaks support pacing.

Your camera. Your choice.

Normal coaching does not automatically record or upload a clip.

01

Process locally

Pose estimation runs on the device during coaching.

02

Review before sharing

A separate short recording is reviewed before the user chooses to share.

03

Keep care connected

Clinic workflows support clinician plans and explicitly shared clips.

Clinic delivery requires configured services and connected accounts. Rebound does not diagnose injuries.

HOW IT WORKS

Built around one shared recovery loop.

A native iPhone experience, shared guided workflows, and local pose processing.

EXPO + REACT NATIVE

iPhone

Recovery home, Coach library, reminders, and native recording.

REACT + TYPESCRIPT

Shared app

Routine state, check-ins, guided sessions, and web presentation.

MEDIAPIPE / SUPABASE

Camera + care

Local pose estimates; optional authenticated Clinic services.

Camera frame

Local video input

Pose landmarks

Joint estimates

Movement checks

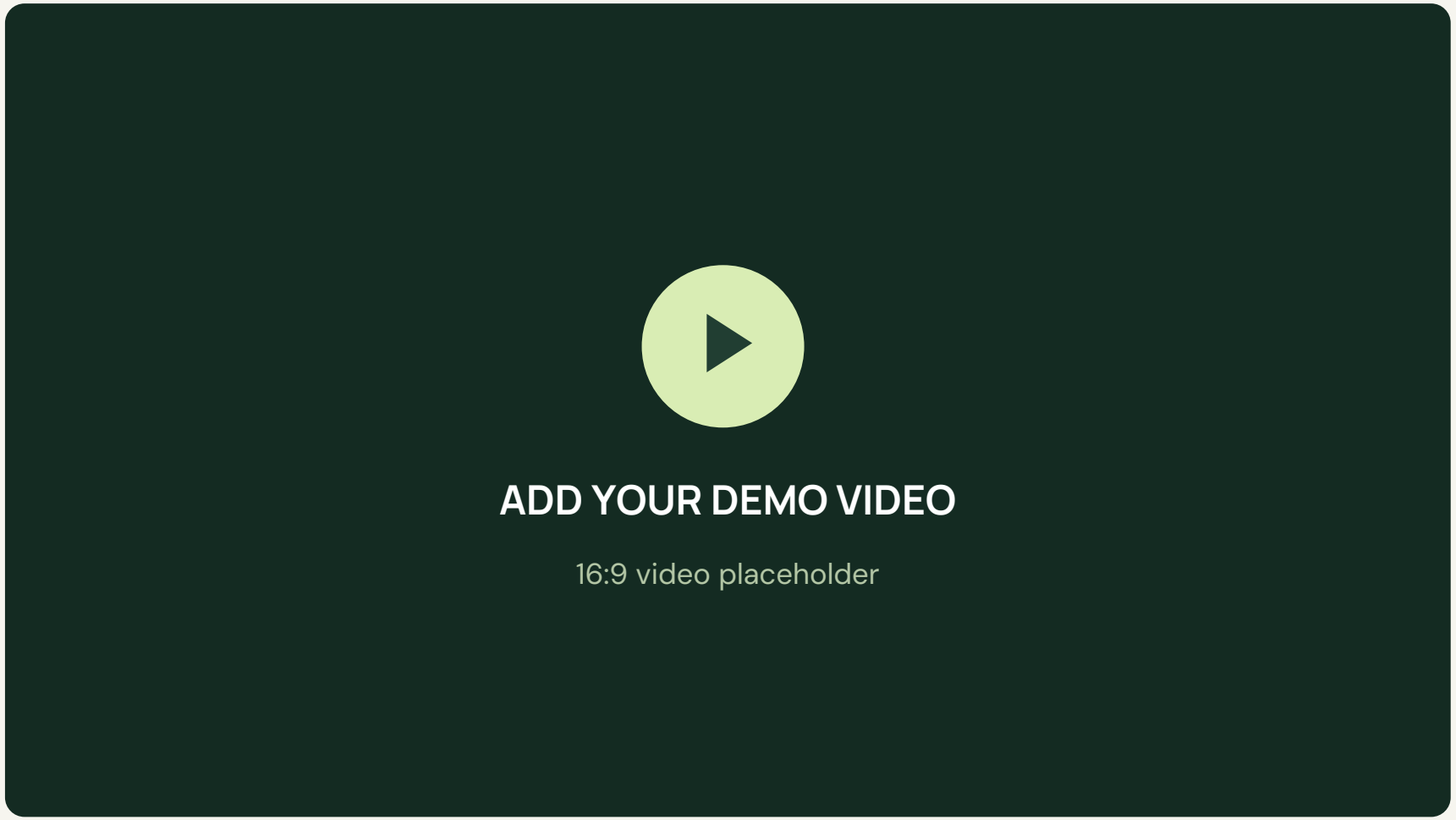
Visibility + timing

Counter or cue

Feedback in the view

VIDEO DEMONSTRATION

See Rebound in action.



SHOW THE LOOP

01 Plan

Show the weekly goal and next routine.

02 Practice

Open a cleared guide and the camera coach.

03 Reflect

Save a session and show the updated record.

A strong prototype. A clear next step.

Keep improving usefulness while measuring what the system can actually do.

ENGINEERING CHECKS

73

browser checks passed

Plus web build, native type-check, and iOS JavaScript export.

01 **Validate on people and iPhones**

Measure rep errors and specific form errors against qualified observers.

02 **Evaluate exercise-specific video data**

Resolve data rights, labels, and band-assisted movement coverage.

03 **Connect a dependable care workflow**

Configure Clinic services, stable HTTPS hosting, and signed native builds.

Engineering checks are not clinical validation; no human-video accuracy percentage is established

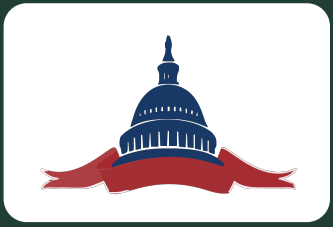
rebound.

A clearer plan.
A steadier routine.

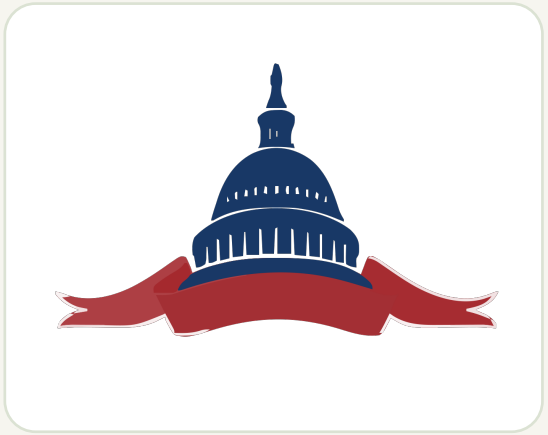
Helping people understand the next step and make room for the routine.

Ritvik Saidapur · Siddharth Senthilkumar

NC-04 / Congressional App Challenge



Sources, identity, and tool credits.



Congressional App Challenge

Official CAC repository asset. Event identification; no endorsement implied.

Rebound identity + app evidence

Current app icon, native components, web UI, and repository branch rebound-mvp.

Movement and pose references

NHS sitting / strength / flexibility / balance guidance; Google MediaPipe Pose documentation.

Logo + typography

CAC: congressionalappchallenge/houseofcode2024 · frontend/src/assets/CAC-logo.svg. Manrope and DM Sans: SIL Open Font License.

Tools + status

React, TypeScript, Expo, React Native, MediaPipe, optional Supabase. OpenAI Codex assisted implementation, debugging, and this presentation.

Screens use example data. Camera slide is a UI test preview, not patient footage. Full source links and import instructions accompany the deck.